BETTINA ANNE SAM

Software Development • AI/ML • Full-Stack Engineering| React • Node.js • MongoDB • SQL • REST APIs • Cloud Technologies • System Design
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PROFESSIONAL SUMMARY

Curiosity-driven Information Technology undergraduate with a strong interest in building practical, technology-driven solutions across full-stack development, machine learning, and interactive applications. Hands-on experience in developing and deploying end-to-end systems using React, Node.js, Python, REST APIs, databases, and modern web technologies. A quick learner and problem solver who enjoys exploring new technologies, transforming ideas into functional products, and continuously expanding technical skills through real-world projects and experimentation

EDUCATION

Francis Xavier Engineering College
B.Tech — Information Technology | 2023 – 2027 | *CGPA: 9.64*

Francis Xavier School (CBSE)
Higher Secondary | 2021 – 2023 | 81.8%

Francis Xavier School (CBSE)
Secondary | 2020 – 2021 | 89.8%

TECHNICAL SKILLS

Frontend: React.js, React Native, JavaScript, TypeScript
Backend & DB: Node.js, Express.js, FastAPI, REST API Integration, MongoDB, MySQL, SQLite
Machine Learning & Data Science: Python, XGBoost, Scikit-learn, Random Forest, Regression, Monte Carlo Simulation, Uncertainty Quantification, Pandas, NumPy
Programming: Java, Python, C, JavaScript, TypeScript, SQL, R
DevOps & Tools: Git, GitHub, Bitbucket, Jira, AWS, Azure, CI/CD, Linux, WebSockets, Appium, Selenium, TestNG

WORK EXPERIENCE

Machine Learning Intern | DRDO – Aeronautical Development Establishment (ADE) *Bangalore | Jun 15 – Jul 31, 2026 (6 weeks Internship)*

- Developed a full-stack, physics-informed machine learning platform for UAV performance analysis and mission planning. Worked on predicting key flight-performance parameters including range, endurance, altitude-related performance, and mission feasibility by integrating machine learning with physics-based computational methods.
- Developed interactive mission-planning capabilities including live map visualization, location marking, route-based analysis, 3D flight Visualization and graphical representation of flight-performance data.
- Engineered a **17-page, 21-endpoint full-stack application** (FastAPI + React/TypeScript, installable as a PWA) covering uncertainty quantification, inverse design search, engine-out failure-mode simulation, sensitivity analysis, and live mission planning; trained on a physics-generated synthetic dataset and **live-deployed** for interactive demonstration.

Assistant Software Engineer | Rotation 5 Company *Aug 2023 – Feb 2026 | Stipend-Based Internship*

- Cross-Platform Application Development: Developed and optimized cross-platform applications, enhancing functionality, performance, and user experience across Android and iOS.
- REST API Integration & Testing: Integrated and validated backend services, ensuring reliable frontend–backend communication and data exchange.
- Software Quality & Automation: Designed and executed 1000+ automated test cases and regression suites using Appium, Selenium, and TestNG, improving release stability and validation efficiency.
- Defect Resolution: Performed structured root-cause analysis and defect management to resolve functional, UI, and performance issues.
- Agile Delivery: Contributed to sprint-based development cycles using Git, Bitbucket, and Jira.

PROJECTS

Digital Labs - Interactive Learning Platform *Node.js · Express.js · MongoDB · React.js · TypeScript · REST APIs* • Framer Motion— *Live Deployed*

- Learning Analytics & Interactive Education: Developed a web platform featuring coding simulations, technology learning modules, and responsive user experiences.

ResQHub - Real-Time Disaster Management *Node.js · Express.js · MongoDB · WebSockets · REST APIs · Leaflet.js — Live Deployed*

- Emergency Response & Real-Time Monitoring: Developed a disaster management platform with live hazard monitoring, geofenced alerts, and emergency reporting capabilities.

Air Sculpt Studio - 3D Creation Platform *React.js · Three.js · MediaPipe · Web APIs — Live Deployed*

- Computer Vision & Gesture Recognition: Built an AI-powered application that converts hand movements into interactive 3D objects using real-time gesture tracking.

Particle Observatory – Interactive Particle Simulation JavaScript • HTML5 Canvas • Web APIs • Interactive Data Visualization — *Live Deployed*

- Built and deployed an interactive particle-observation platform for visualizing dynamic particle behavior through real-time simulations, configurable parameters, and responsive visual interactions.

Ocean Depths Dive – Interactive Exploration Platform *React.js • TypeScript • Tailwind CSS • Framer Motion • Three.js — Live Deployed*

- Designed and deployed an immersive ocean-exploration experience combining interactive interfaces, animated transitions, 3D elements, and responsive web performance.

ACHIEVEMENTS

- 1000+ Problems Solved (SkillRack): Algorithmic problem-solving in C & Python - 2+ years of consistent practice
- HackerRank Certified: SQL, Java, C, Python, JavaScript - 5 domains verified
- NEC National Symposium: 1st Place - back-end architecture presented to national-level technical panel
- Technoxian World Cup 2024: National level, Noida - systems design under time-constrained competition

CERTIFICATIONS & KEY COURSEWORK

- NPTEL: Internet of Things - Top 2% (92%) | Joy of Computing Elite + Gold Medal • Udemy: Java Bootcamp (66 hrs) | C Programming (44 hrs) | Appium Automation (26 hrs)
- Great Learning: Cloud Computing | DevOps | Linux | Cybersecurity | Data Analysis • LinkedIn: Gen AI Essentials (Microsoft & Google) | Ethics in Gen AI | Copilot
- AICTE STTP: Java Full Stack with React JS & AI - National Program 2024
- Core Coursework: Web Technologies • Data Structures & Algorithms • OOP • DBMS • Software Engineering • Operating Systems • Computer Networks